

Apply filters to SQL queries

Project description

In this scenario, I need to obtain specific information about employees, their machines, and the departments they belong to from the database.

My team needs data to investigate potential security issues and to update computers.

I'm responsible for filtering the required information from the database.

Here's how I'll do this task: **First**, I'll retrieve all failed login attempts after business hours. **Second**, I'll retrieve all login attempts that occurred on specific dates. **Third**, I'll retrieve logins that didn't originate in Mexico. **Fourth**, I'll retrieve information about certain employees in the Marketing department. **Fifth**, I'll retrieve information about employees in the Finance or the Sales department. **Finally**, I'll obtain information about employees who are not in the Information Technology department.

Retrieve after hours failed login attempts

My team is investigating failed login attempts that were made after business hours. I want to retrieve this information from the login activity. I'll identify all unsuccessful attempts after 18:00.

```

MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_time > '18:00' AND success = False;
+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address |
| success |
+-----+-----+-----+-----+-----+-----+
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.12 |
| 0 |
| 18 | pwashing | 2022-05-11 | 19:28:50 | US | 192.168.66.142 |
| 0 |
| 20 | tshah | 2022-05-12 | 18:56:36 | MEXICO | 192.168.109.50 |
| 0 |
| 28 | aestrada | 2022-05-09 | 19:28:12 | MEXICO | 192.168.27.57 |
| 0 |
| 34 | drosas | 2022-05-11 | 21:02:04 | US | 192.168.45.93 |
| 0 |
| 42 | cgriffin | 2022-05-09 | 23:04:05 | US | 192.168.4.157 |
| 0 |
| 52 | cjackson | 2022-05-10 | 22:07:07 | CAN | 192.168.58.57 |
| 0 |
| 69 | wjaffrey | 2022-05-11 | 19:55:15 | USA | 192.168.100.17 |
| 0 |
| 82 | abernard | 2022-05-12 | 23:38:46 | MEX | 192.168.234.49 |
| 0 |
| 87 | apatel | 2022-05-08 | 22:38:31 | CANADA | 192.168.132.153 |
| 0 |
| 96 | ivelasco | 2022-05-09 | 22:36:36 | CAN | 192.168.84.194 |
| 0 |
| 104 | asundara | 2022-05-11 | 18:38:07 | US | 192.168.96.200 |
| 0 |
| 107 | bisles | 2022-05-12 | 20:25:57 | USA | 192.168.116.187 |
| 0 |
| 111 | aestrada | 2022-05-10 | 22:00:26 | MEXICO | 192.168.76.27 |
| 0 |
| 127 | abellmas | 2022-05-09 | 21:20:51 | CANADA | 192.168.70.122 |
| 0 |

```

Retrieve login attempts on specific dates

My team is investigating a suspicious event that occurred on '2022-05-09'. I want to retrieve all login attempts that occurred on this day and the day before ('2022-05-08').

MariaDB [organization]> SELECT *

-> FROM log_in_attempts

-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
39	yappiah	2022-05-09	07:56:40	MEXICO	192.168.57.115	1

Retrieve login attempts outside of Mexico

Now, my team is investigating logins that did not originate in Mexico, and I need to find this information. Note that the country field includes entries with 'MEX' and 'MEXICO'. You should use the NOT and LIKE operators and the matching pattern 'MEX%'.

```
MariaDB [organization]> SELECT *
  -> FROM log_in_attempts
  -> WHERE IN country LIKE 'MEX%';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'IN country LIKE 'MEX%'' at line 3
MariaDB [organization]> SELECT *
  -> FROM log_in_attempts
  -> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243
8	bisles	2022-05-08	01:30:17	US	192.168.119.173
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158
13	mrach	2022-05-11	09:29:34	USA	192.168.246.135
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99

Retrieve employees in Marketing

Your team is updating employee machines, and you need to obtain the information about employees in the 'Marketing' department who are located in all offices in the East building (such as 'East-170' or 'East-320').

```
MariaDB [organization]> SELECT *
-> FROM employees;
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-27
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1006	g329h357i597	alevitsk	Information Technology	East-320
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1012	m756n668o146	nmason	Information Technology	North-160
1013	n205o559p243	zbernal	Information Technology	South-229
1014	NULL	asundara	Information Technology	West-219
1015	p611q262r945	jsoto	Finance	North-271

Retrieve employees in Finance or Sales

Now, my team needs to perform a different update to the computers of all employees in the Finance or the Sales department, and you need to locate information on these employees.

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1029	d336e475f676	ivelasco	Finance	East-156
1035	j236k303l245	bisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929q222r778	cgriffin	Sales	North-208
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1057	f370g535h632	mscott	Sales	South-270
1062	k367l639m697	redwards	Finance	North-180
1063	l686m140n569	lpope	Sales	East-226
1066	o678p794q957	ttyrell	Sales	Central-444
1069	NULL	jpark	Finance	East-110
1071	t244u829v723	zdutchma	Sales	West-348
1072	u905v920w694	esmith	Sales	East-421
1076	y347z204a710	fgarcia	Finance	Central-270
1078	a667b270c984	sharley	Sales	North-418
1081	d647e310f618	qcorbit	Finance	South-290

Retrieve all employees not in IT

My team needs to make one more update. This update was already made to employee computers in the Information Technology department. The team needs information about employees who are not in that department. I should use the NOT operator to identify these employees.

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE NOT department LIKE 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1016	q793r736s288	sbaelish	Human Resources	North-229
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1020	u899v381w363	arutley	Marketing	South-351
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1026	a998b568c863	apatel	Human Resources	West-320
1027	b806c503d354	mrah	Marketing	West-246
1028	c603d749e374	aestrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156
1030	e391f189g913	mabadi	Marketing	West-375
1031	f419g188h578	dkot	Marketing	West-408
1034	i679j565k940	bsand	Human Resources	East-484
1035	j236k303l245	bisles	Sales	South-171
1036	k550l533m205	rjensen	Marketing	Central-239
1038	m873n636o225	btang	Human Resources	Central-260
1039	n253o917p623	cjackson	Sales	East-378
1040	o783p832q294	dtarly	Human Resources	East-237
1041	p929q222r778	cgriffin	Sales	North-208
1042	q175r338s833	acook	Human Resources	West-381

Summary

I mastered how to write and execute SQL queries to extract targeted information from a relational database. Specifically, I learned to implement conditional logic using **AND**,

OR, and **NOT** operators within WHERE clauses to filter complex datasets and isolate precise records based on specific security and operational criteria.